Project Design Phase-I
Solution Architecture

Date	09 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Solution Architecture:

- Solution architecture is a structured framework that defines how different components of a system interact to solve a specific business problem.
- In this project, the solution architecture is designed to analyze housing market data and visualize important trends using Tableau dashboards integrated into a web application developed using Flask.
- The architecture connects raw housing datasets with interactive visual insights by combining data preprocessing, data visualization, and web deployment.
- The main goals of this architecture are:
- • To collect housing market datasets from reliable sources.
  • To perform data cleaning and preprocessing.
  • To analyze the relationship between housing features and sale prices.
  • To generate interactive dashboards and stories using Tableau.
  • To integrate Tableau dashboards into a web application using Flask.
  • To help users explore housing trends through an interactive web interface.
- This architecture allows users to understand housing price patterns, property characteristics, and market behavior through visual analytics.
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- **2. Components of the Solution Architecture**

  The proposed architecture consists of five major layers:
  1. Data Source Layer
  This layer represents the origin of the housing dataset.
  The dataset contains housing information such as:
  • Sale Price
  • Number of Bedrooms
  • Number of Bathrooms
  • Lot Area
  • Year Built
  • Floors
  • Renovation Year
  The dataset is typically stored in CSV or Excel format.

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  2. Data Processing Layer
  This layer performs preprocessing and cleaning of the raw data.
  Data processing tasks include:
  • Removing missing values
  • Handling duplicate records
  • Converting data types

• Standardizing column names
• Creating calculated fields
Data preprocessing ensures that the dataset is clean and ready for analysis.

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3. Visualization Layer
In this layer, the processed dataset is connected to Tableau Desktop.
Tableau transforms the dataset into interactive visualizations such as:
• Bar charts
• KPI chart
• Pie chart
• Histograms
These visualizations help identify trends in housing prices and property features.

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4. Application Layer
The application layer integrates the Tableau dashboards into a web interface.
This is implemented using Flask, a lightweight Python web framework.
The Flask application performs the following tasks:
• Handles routing of web pages
• Loads HTML templates
• Embeds Tableau dashboards
• Displays visualizations in the browser
This allows users to interact with dashboards through a web application.

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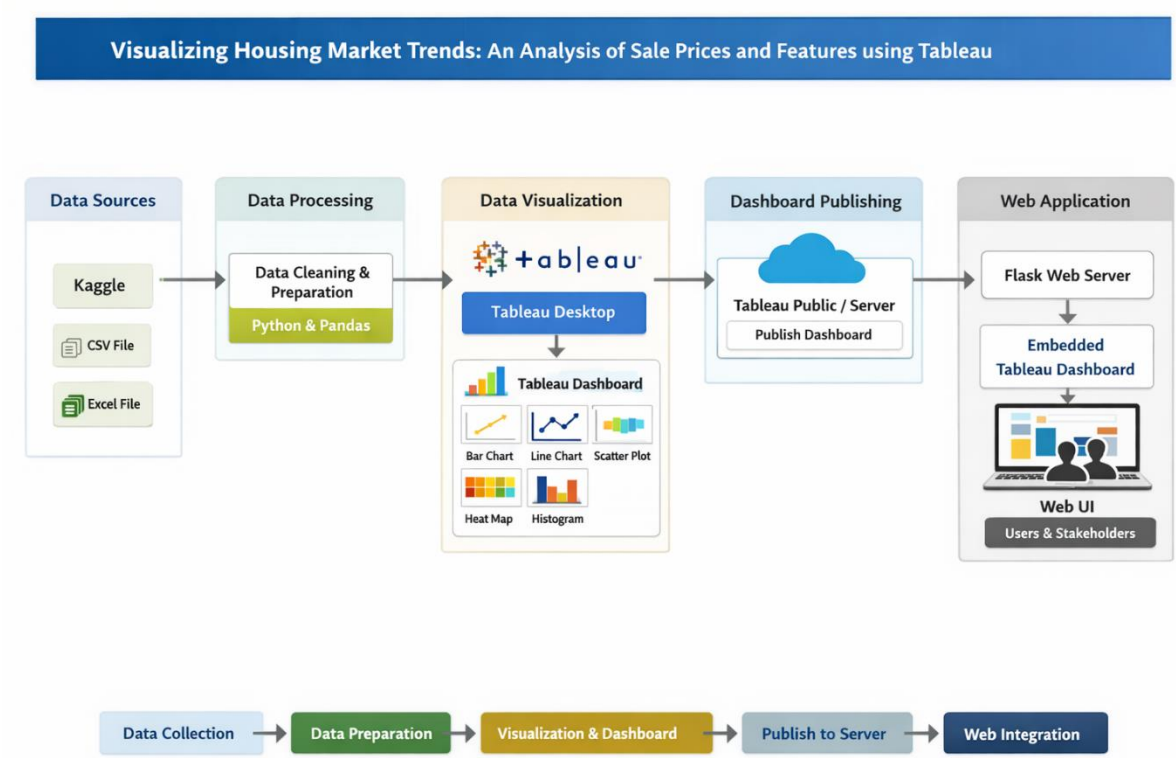
5. User Interface Layer
The user interface layer represents the front-end of the web application.
The interface is developed using:
• HTML
• CSS
• Bootstrap
Users can navigate through sections such as:
• Home
• About
• Dashboard
• Story
The UI allows users to explore the housing market insights visually.

Example - Solution Architecture Diagram:



4. Data Flow Architecture

The data flows through multiple stages from raw dataset to final visualization.

Step 1: Data Collection

Housing datasets are collected from sources such as:

- Kaggle
- UCI Machine Learning Repository
- Government housing databases

The dataset typically includes:

- Sale Price
- Lot Area
- Bedrooms
- Bathrooms
- Year Built
- floors

The data is stored in CSV or Excel format.

Step 2: Data Preparation

Before visualization, the dataset undergoes data preprocessing.

Data preparation tasks include:

- Removing missing values
- Handling outliers
- Converting data types
- Renaming columns
- Feature engineering

Example calculated fields:

- **House Age**
  Current Year – Year Built
- **Price per Square Foot**
  Sale Price / Lot Area
- **Price Category**
- • Low
  • Medium
  • High
- These calculations improve the quality of analysis.

Step 3: Data Modeling in Tableau

The cleaned dataset is connected to Tableau Desktop.

Tableau automatically categorizes data into:

Dimensions
Example:

- Neighborhood
- House Type

Measures
Example:

- Sale Price
- Lot Area

Calculated fields are created to derive additional insights.

Step 4: Data Visualization

Multiple interactive visualizations are created in Tableau.

Examples include:

Bar Chart
Average Sale Price by Neighborhood

Line Chart
Sale Price Trend by Year

Scatter Plot
Lot Area vs Sale Price

Heat Map
Neighborhood vs Average Price

Histogram
Distribution of Housing Prices

These visualizations help identify patterns and trends in housing markets.

Step 5: Dashboard Development

The visualizations are combined into a single interactive dashboard.

Dashboard features include:

- Interactive filters
- Tooltips
- Drill-down analysis
- Responsive layout

Users can filter data based on:

- Location
- Price range
- Number of bedrooms
- Year built

Step 6: Storyboard Creation

Tableau Story is used to present insights sequentially.

Example story flow:

1. Overview of housing market
2. Price comparison by neighborhood
3. Effect of house size on price
4. Price trend over time
5. Key insights and conclusions

This helps stakeholders understand the results easily.

Step 7: Dashboard Publishing

The dashboard is published using:

- Tableau Public
- Tableau Server
- Tableau Online

After publishing, Tableau generates an embed link.

Step 8: Web Integration using Flask

A Flask web application is created to embed the Tableau dashboard.

Flask handles:

- Web routing
- HTML page rendering
- Dashboard embedding

Example architecture:

Flask Server → HTML Page → Embedded Tableau Dashboard

This allows users to access the dashboard through a web interface.

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5. Technology Stack

Layer	Technology Used
Data Storage	CSV / Excel
Data Processing	Python, Pandas
Visualization	Tableau Desktop

Layer	Technology Used
Dashboard Hosting	Tableau Public
Web Integration	Flask
Frontend	HTML, CSS

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6. Performance Optimization

To ensure smooth performance, the following methods are used:

- Data extracts instead of live queries
- Optimized calculated fields
- Limited filters and visualizations
- Efficient data aggregation

These techniques improve dashboard responsiveness.

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7. Benefits of the Solution Architecture

The proposed architecture provides several advantages:

- Scalable data analysis
- Interactive visual exploration
- Easy deployment through web interface
- Clear understanding of housing market trends
- Support for decision-making in real estate markets

The architecture allows stakeholders to analyze housing price patterns and property features effectively.